

Solutions to JEE Main - 11 | JEE-2022

PHYSICS

SECTION-1

1.(D) $\vec{E} = \frac{\sigma}{2\epsilon_0}(-\hat{i} - \hat{j})$

$$\vec{\Delta r} = (\hat{i} - \hat{j} + \hat{k})d$$

$$W = \vec{QE} \cdot \vec{\Delta r} = -\frac{\sigma Qd}{2\epsilon_0} + \frac{\sigma Qd}{2\epsilon_0} = 0$$

2.(C) Say final charges are q_1 & q_2

$$q_1 + q_2 = Q$$

$$\frac{Kq_1}{R} - \frac{Kq_2}{R} = E \Rightarrow q_1 - q_2 = \frac{Q}{2}$$

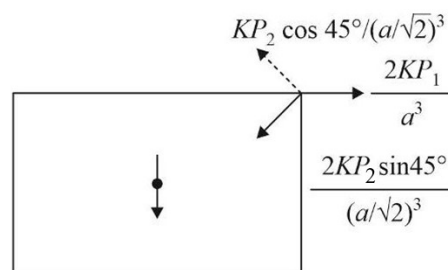
$$\Rightarrow q_1 = \frac{3Q}{4} \quad q_2 = \frac{Q}{4}$$

$$\begin{aligned} \text{Heat loss} &= \left[\frac{K(2Q)^2}{2R} - \frac{K(3Q/4)^2}{2R} \right] + \left[\frac{K(Q)^2}{2R} - \frac{K(Q/4)^2}{2R} \right] - E \times \left(2Q - \frac{3Q}{4} \right) \\ &= \frac{4KQ^2}{2R} - \frac{9KQ^2}{32R} + \frac{KQ^2}{2R} - \frac{KQ^2}{32R} - \frac{KQ}{2R} \times \frac{5Q}{4} \\ &= \frac{5KQ^2}{2R} - \frac{30KQ^2}{32R} = \frac{50KQ^2}{32R} = \frac{25Q^2}{64\pi\epsilon_0 R} \end{aligned}$$

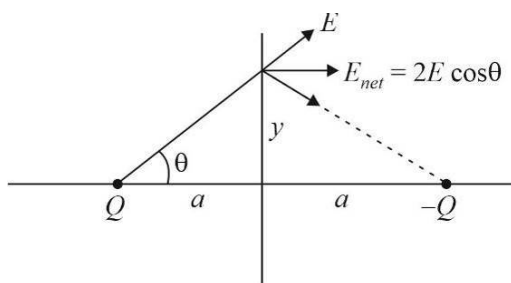
3.(A) $\frac{2KP_1}{a^3} = \frac{1}{\sqrt{2}} \frac{KP_2(\cos 45^\circ)2\sqrt{2}}{a^3} + \frac{1}{\sqrt{2}} \frac{2KP_2(\sin 45^\circ)2\sqrt{2}}{a^3}$

$$2P_1 = \frac{2P_2 + 4P_2}{\sqrt{2}}$$

$$\frac{P_1}{P_2} = \frac{6}{\sqrt{2} \times 2} = \frac{3}{\sqrt{2}}$$



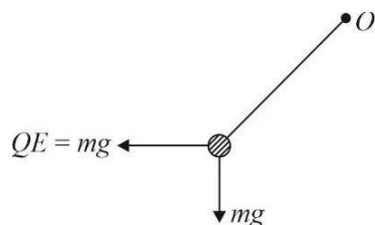
4.(A)



$$= \frac{2KQ}{y^2 + a^2} \frac{a}{(y^2 + a^2)^{1/2}} = \frac{2KQa}{(y^2 + a^2)^{3/2}}$$

5.(C) $\vec{E} = 3\hat{i} - 2\hat{j}$

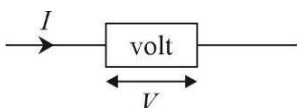
6.(B)



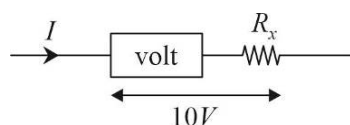
$$g_{\text{eff}} = \sqrt{2}g$$

$$v_{\text{min}} = \sqrt{5\sqrt{2}g\ell}$$

7.(C)



$$IR = V$$



$$I(R + R_V) = 10V$$

$$\frac{R + R_V}{R} = 10$$

$$R_V = 9R = 180\Omega$$

8.(C) $i = \frac{E}{2R} + \frac{E}{R} e^{-t/RC}$

$$\Rightarrow i = \frac{E}{2R} + \frac{E}{R} = \frac{E}{R} \left[\frac{e+2}{2e} \right]$$

9.(C) $\frac{k(-2q)}{\sqrt{x^2 + y^2}} + \frac{kq}{\sqrt{(x-4)^2 + y^2}} = 0$

$$4[(x-4)^2 + y^2] = x^2 + y^2$$

$$4[x^2 - 8x + 16 + y^2] = x^2 + y^2$$

$$3x^2 + 3y^2 - 32x + 64 = 0$$

Circle with center $\left(\frac{16}{3}, 0\right)$

10.(B) Potential gradient $= \frac{20V}{150\text{cm}}$

$$\frac{20}{150} \times x = \left(\frac{8}{1} - \frac{6}{2}\right) \frac{2}{3}$$

$$\frac{2}{15}x = \frac{10}{3} \Rightarrow x = \frac{150}{6} = \frac{50}{2} = 25\text{cm}$$

11.(C) Charge on the inner-surface will be zero.

12.(C) $\frac{H}{V} = \frac{i^2 R}{\text{volume}} = E^2 \sigma$

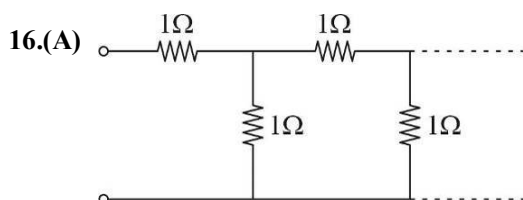
13.(D) Higher the value of k , degree of polarization is higher.

14.(D) Value of r has no effect on position of null point as it doesn't effect potential difference between A & J or potentiometer wire.

15.(D) Say current $= i_0$

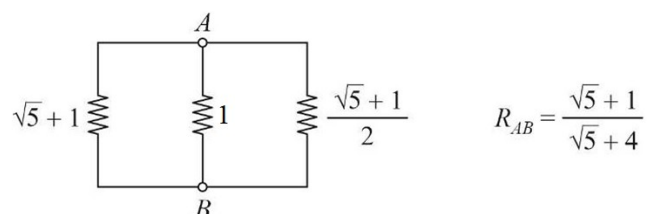
$$V_0 - V_x = i_0 \left[\int_0^x \frac{\rho_0 e^{-x/L} dx}{A} \right]$$

$$V_0 - V_x = \frac{L\rho_0 i_0}{A} [1 - e^{-x/L}] \Rightarrow V_x = V_0 + \frac{\rho_0 i_0 L}{A} e^{-x/L} - \frac{\rho_0 i_0 L}{A}$$



$$x = \frac{x}{x+1} + 1 \Rightarrow x^2 + 1 = x^2 + x \Rightarrow x^2 - x - 1 = 0 \Rightarrow x = \frac{1+\sqrt{5}}{2}$$

Circuit becomes



17.(A) Initially $q_i = E \times \frac{2C}{3}$

Finally $q_f = E \times \frac{2KC}{K+2}$

Charge supplied by battery $= \frac{CE}{3} = q_f - q_i \Rightarrow K = 2$

18.(D) $q = CV$

$$\frac{V_{key\ open}}{V_{key\ closed}} = \frac{6V}{10V} = \frac{6}{10} = \frac{3}{5}$$

19.(C) Rate of energy transfer to $C = iV_C = \frac{V - V_C}{R} \times V_C$

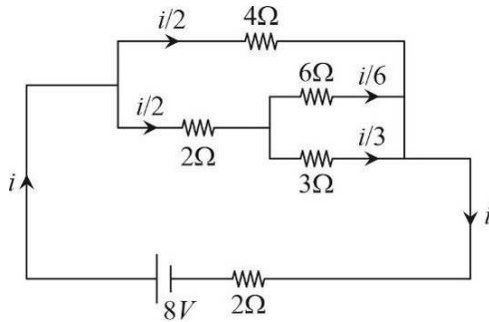
Rate is maximum when $V_C = \frac{V}{2}$

Energy stored in capacitor $\frac{CV^2}{8}$

20.(A) $V_+ - V_C = Ed + \frac{E}{10} \times \frac{d}{2}$; $V_+ - V_- = Ed + \frac{Ed}{10} = \frac{Q_0}{C} \Rightarrow V_+ - V_C = \frac{21Q_0}{22C}$

SECTION-2

21.(1.33)

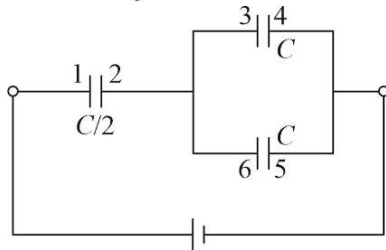
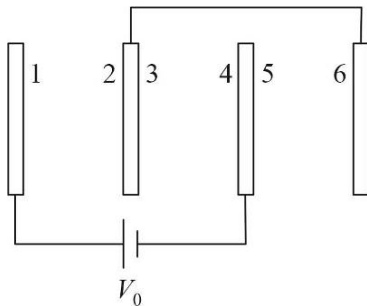


$$\text{Reading in Ammeter} = \frac{i}{2} + \frac{i}{6} = \frac{2i}{3} = \frac{4}{3} = 1.33 \text{ A}$$

22.(6.28)

$$\begin{aligned} \phi &= E_0 \left(\frac{3}{4} \right) \pi (1)^2 + E_0 \left(\frac{1}{4} \right) \pi (1)^2 \\ &= E_0 \pi (1)^2 \\ &= 2 \times 3.14 = 6.28 \end{aligned}$$

23.(4)



$$C = \frac{A \epsilon_0}{d} \quad C_{eq} = \frac{2C \times \frac{C}{2}}{2C + \frac{C}{2}} = \frac{2C}{5}$$

$$q = \frac{2C}{5} \times 10 = 4\mu C$$

24.(2400) $r = \sqrt{9 \times 4} = 6\Omega$; $P_{\max} = \frac{\epsilon_0}{4r} = \frac{(240)^2}{24} = 2400 \text{ watts}$

25.(18) Electric field inside cavity $= \frac{\rho \left(2R - \frac{3R}{2} \right)}{3\epsilon_0} = \frac{\rho R}{6\epsilon_0}$; $V_A - V_B = \frac{\rho R}{6\epsilon_0} \times 3R = \frac{\rho R^2}{2\epsilon_0} = 8 \times \frac{9}{4} = 18V$

Chemistry

SECTION-1

1.(D) Melting point is very low for molecular solids.

2.(B) Informative

3.(C) Solution of methanol and ethanol is almost ideal.

$$4.(B) \frac{Y_A}{Y_B} = \frac{X_A}{X_B} \cdot \frac{P_A^\circ}{P_B^\circ} \Rightarrow \frac{0.5}{0.5} = \frac{0.25}{0.75} \times \frac{P_A^\circ}{P_B^\circ} \Rightarrow \frac{P_A^\circ}{P_B^\circ} = \frac{3}{1}$$

5.(B) Check options

$$6.(C) r = k[\text{NO}]^2[\text{O}_2]$$

On tripling the volume, the concentration becomes

$\frac{1}{3}$ times and hence, rate becomes $\frac{1}{27}$ times.

$$7.(C) \Delta T_b = K_b \cdot m \Rightarrow 0.26 = 0.52 \times \frac{w_1 / 150}{w_2 / 1000} \Rightarrow \frac{w_1}{w_2} = \frac{3}{40}$$

$$\therefore \text{Mass percent of solute} = \frac{w_1}{w_1 + w_2} \times 100\% = 6.98\%$$

8.(A) Informative

$$9.(D) \frac{[C]}{[B]} = \frac{k_2}{k_1}$$

10.(D) Emission of each α particle decreases atomic number by 2 and mass number by 4. Emission of each β particle increases atomic number by 1.

$$11.(B) \Delta T_f + \Delta T_b = (K_f + K_b) \cdot m$$

$$4.76 = (1.86 + 0.52) \times \frac{w / 342}{100 / 1000} \Rightarrow w = 68.4 \text{ gm}$$

$$12.(B) \frac{r_{\text{Rb}^+}}{r_{\text{I}^-}} = \frac{1.46}{2.16} = 0.676$$

$\Rightarrow$ NaCl structure

13.(A) Arsenic belongs to group 15

$$14.(B) Y_A = \frac{P_A}{P_{\text{total}}} = \frac{X_A \cdot P_A^\circ}{P_{\text{total}}}$$

$$\frac{1}{X_A} = \frac{1}{Y_A} \cdot P_A^\circ \cdot \left(\frac{Y_A}{P_A^\circ} + \frac{(1 - Y_A)}{P_B^\circ} \right)$$

$$= \frac{1}{Y_A} \cdot \frac{P_A^\circ}{P_B^\circ} + \frac{P_B^\circ - P_A^\circ}{P_B^\circ}$$

15.(D) Reaction will complete before 2 hours

16.(C) $E_a \uparrow, r \downarrow$

17.(A) O_2 is more soluble in water than H_2

18.(C) Let x atoms are in +2 state.

$$x(+2) + (1.92 - x)(+1) + 1.00(-2) = 0$$

$$\Rightarrow x = 0.08$$

$$\therefore \frac{N_{Cu^{2+}}}{N_{Cu^{+}}} = \frac{0.08}{1.92 - 0.08} = \frac{1}{23}$$

19.(D) $\Delta H = E_{af} - E_{ab} \Rightarrow y - x = E_{af} - z \Rightarrow E_{af} = y - x + z$

20.(B) $r_{rxn} = -\frac{1}{2} \cdot \frac{d[NO_2]}{dt} = k_1[NO_2]^2 - k_2[N_2O_4]$

$\therefore$ Rate of disappearance of NO_2 is given by,

$$-\frac{d[NO_2]}{dt} = 2k_1[NO_2]^2 - 2k_2[N_2O_4]$$

SECTION-2

21.(5) $d = \frac{Z \cdot M}{N_A \cdot V}$

$$= \frac{4 \times 58.5}{6 \times 10^{23} \times (500 \times 10^{-10})^3} = 3.12 \text{ g/ml}$$

$$\therefore \text{Percentage vacancy} = \frac{3.12 - 2.964}{3.12} \times 100 = 5\%$$

22.(3) $\pi = \frac{\pi_1 V_1 + \pi_2 V_2}{(V_1 + V_2)} = \frac{2.4 \times 2V + 4.2 \times V}{2V + V} = 3 \text{ atm}$

23.(6) $\frac{k_{BrO^-}}{3} = \frac{k_{BrO_3^-}}{1} = \frac{k_{Br^-}}{2}$

$$\therefore k_{BrO_3^-} = a = \frac{0.06}{3} = 0.02 \text{ M}^{-1} \text{ s}^{-1}$$

$$\text{and } k_{Br^-} = b = \frac{2}{3} \times 0.06 = 0.04 \text{ M}^{-1} \text{ s}^{-1}$$

24.(6) $Z_{ZnS} = 4$ but due to Schottky defect, $Z_{ZnS} = 3$

25.(1) $\frac{k_I}{k_{II}} = \frac{A_I \cdot e^{-E_{aI}/RT}}{A_{II} \cdot e^{-E_{aII}/RT}} = \frac{A_I}{A_{II}} \cdot e^{-(E_{aI} - E_{aII})/RT}$

$$= 100 \times e^{-4.606 \times 10^3 / 2 \times 500} = 1$$

Mathematics

SECTION-1

$$1.(A) \quad \text{Let } I = \int \frac{d(x^2 + 1)}{\sqrt{(x^2 + 2)}} = \int \frac{d(x^2 + 1)}{\sqrt{(x^2 + 1) + 1}}$$

$$\text{Put } x^2 + 1 = t \Rightarrow d(x^2 + 1) = dt$$

$$\therefore \quad I \int \frac{dt}{\sqrt{t+1}} = 2\sqrt{t+1} + c$$

$$2\sqrt{(x^2 + 2)} + c$$

$$2.(D) \quad \int \left(\frac{\sec x}{\sec x - \tan x} \right) dx$$

$$= \int \frac{\sec x(\sec x + \tan x)}{\sec^2 x - \tan^2 x} dx$$

$$= \int \sec x(\sec x + \tan x) dx$$

$$= \int \sec^2 x dx + \int \sec x \tan x dx$$

$$= \tan x + \sec x + C$$

$$3.(D) \quad f(1^-) = f(1^+) = f(1)$$

$$\Rightarrow \quad A + B = -2 + \frac{\pi}{4}$$

$$\text{Also } f'(1^-) = f'(1^+) \Rightarrow 3A = \frac{1}{2} \Rightarrow A = \frac{1}{6}$$

$$\text{and } B = \frac{\pi}{4} - \frac{13}{6} = \frac{3\pi - 26}{12}$$

$$A - B = \frac{1}{6} - \frac{\pi}{4} + \frac{13}{6} = \frac{28 - 3\pi}{12}$$

$$\frac{A}{B} = \frac{2}{3\pi - 26}$$

$$4.(A) \quad f(x) = \log_e \left\{ \frac{u(x)}{v(x)} \right\}$$

$$= \log_e u(x) - \log_e v(x)$$

$$f'(x) = \frac{u'(x)}{u(x)} - \frac{v'(x)}{v(x)}$$

$$f'(2) = \frac{u'(2)}{u(2)} - \frac{v'(2)}{v(2)} = \frac{4}{2} - \frac{2}{1} = 2 - 2 = 0$$

5.(A) $2yy' = (2-x)^2 - 2x(2-x)$

$$\left. \frac{dy}{dx} \right|_{(1,1)} = \frac{-1}{2},$$

Equation of tangent $x + 2y = 3$

Again intersect at $P, \left(\frac{3-x}{2} \right)^2 = x(2-x)^2$

1, 1, α are roots of equation $4x^3 - 17x^2 + 22x - 9 = 0$

$$\alpha = \frac{17}{4} - 2 = \frac{9}{4}, \beta = \frac{3}{8}, (\alpha, \beta) = \left(\frac{9}{4}, \frac{3}{8} \right)$$

6.(C) Put $\tan x = t$

$$I = \int \frac{(1+t^2)(1+t^2)^2}{t^4} \frac{dt}{1+t^2}$$

$$I = \int \frac{(1+t^2)^2}{t^4} dt = \int \frac{1+t^4+2t^2}{t^4} dt = \int \left(\frac{1}{t^4} + 1 + \frac{2}{t^2} \right) dt$$

$$= \frac{1}{-3t^3} + t + \frac{2}{-t} + c$$

$$I = \frac{1}{-3 \tan^3 x} + \tan x - \frac{2}{\tan x} + c \Rightarrow c = 0$$

$$\text{So, } f(x) = \frac{1}{-3 \tan^3 x} + \tan x - \frac{2}{\tan x}$$

$$\Rightarrow f\left(\frac{\pi}{4}\right) = -\frac{1}{3} + 1 - 2 = -\frac{4}{3}$$

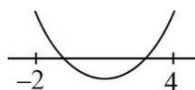
7.(A) $f'(x) = 3x^2 - 3\alpha(2x) + 3(\alpha^2 - 1) = 3x^2 - 6\alpha x + 3(\alpha^2 - 1)$ since, extrema lie in $(-2, 4)$

So roots of $f'(x)$ lie in $(-2, 4)$

$$f'(x) = 3(x - (\alpha - 1))(x - (\alpha + 1))$$

$$\Rightarrow -2 < \alpha - 1 \text{ and } \alpha + 1 < 4$$

$$\Rightarrow -1 < \alpha < 3$$



8.(D) $f(x) = \begin{cases} (x+1)2^{-2/x} & , x > 0 \\ 0 & , x = 0 \\ x+1 & , x < 0 \end{cases}$

$$\text{RHL} = \lim_{x \rightarrow 0^+} (x+1)2^{-2/x} = 0$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} (x+1) = 1$$

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

$$9.(B) \quad \lim_{x \rightarrow e} \frac{\ln x - 1}{x - e} \Rightarrow \lim_{x \rightarrow e} \frac{1}{x} = \frac{1}{e}$$

$$\therefore k = \frac{1}{e}$$

10.(C) $f(x) = |x| \sin x + |(x - \pi)(x + \pi)| \cos x$ points to be checked $x = 0, \pi, -\pi$ at $x = 0$ function is differentiable and not differentiable at $x = \pi, -\pi$

$$11.(B) \quad f'(x) = x^2 - (m - 3)x + m \geq 0$$

$$D \leq 0$$

$$(m - 3)^2 - 4m \leq 0$$

$$m \in [1, 9]$$

$$12.(B) \quad \int \frac{5x^8 + 7x^6}{x^{14} \left(2 + \frac{1}{x^7} + \frac{1}{x^5} \right)^2} dx = \int \frac{\frac{5}{x^6} + \frac{7}{x^8}}{\left(2 + \frac{1}{x^7} + \frac{1}{x^5} \right)^2} dx$$

$$\text{Put } 2 + \frac{1}{x^7} + \frac{1}{x^5} = t$$

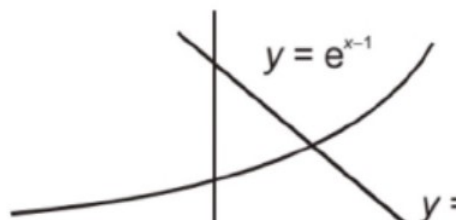
$$\Rightarrow f(x) = \left(\frac{x^7}{(2x^7 + x^2 + 1)} \right) + C$$

$$\therefore f(0) = 0 \Rightarrow C = 0 \Rightarrow f(x) = \frac{x^7}{(2x^7 + x^2 + 1)}$$

$$\therefore f(1) = \frac{1}{4}$$

13.(A) The given equation can be written in the form $e^{x-1} = 2 - x$

Let us draw the graphs of $y = e^{x-1}$ and $y = 2 - x$ and the point of intersection of the two curves gives the number of real roots of the equation.



From the graph of the two curves it follows that the given equation has one real root.

$$14.(B) \quad \int \frac{x e^{\tan^{-1} x^2}}{x^4 + 1} dx$$

$$= \frac{1}{2} \int e^{\tan^{-1} x^2} \cdot \frac{2x}{1 + (x^2)^2} dx$$

$$= \frac{1}{2} \int e^t dt = \frac{1}{2} e^t + c$$

$$= \frac{1}{2} e^{\tan^{-1} x^2} + c$$

Substituting $\tan^{-1} x^2 = t$

$$\frac{2x}{1 + x^4} dx = dt$$

$$15.(D) \quad f(x) = \lim_{t \rightarrow 0} \tan^{-1} \left(\frac{(e^{xt} - 1)}{tx} \times x \right) = \tan^{-1} x$$

$$\lim_{x \rightarrow 0} \frac{f(x) - x}{x^3} = \lim_{x \rightarrow 0} \frac{\tan^{-1} x - x}{x^3} = -\frac{1}{3}$$

$$16.(B) \quad y = \frac{x^{12} + x^6 + 1}{x^6 + x^3 + 1} = \frac{x^{12} + 2x^6 + 1 - x^6}{x^6 + x^3 + 1}$$

$$= \frac{(x^6 + 1)^2 - (x^3)^2}{x^6 + x^3 + 1} \Rightarrow y = x^6 - x^3 + 1$$

$$\Rightarrow \frac{dy}{dx} = 6x^5 - 3x^2$$

$$\therefore a = 6, b = -3$$

$$a + b = 6 - 3 = 3.$$

$$17.(A) \quad g(x) = f(x) + f(2 - x)$$

$$g'(x) = f'(x) - f'(2 - x)$$

For g to be increase

$$x > 2 - x$$

$$x > 1$$

$$18.(C) \quad \text{Let } n = \frac{1}{x}$$

$$l = \lim_{x \rightarrow 0} \frac{e(1-x)^{1/x} - 1}{(1/x)^\alpha} = \lim_{x \rightarrow 0} \frac{e \cdot e^{\frac{\ln(1-x)}{x}} - 1}{x^{-\alpha}}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\frac{\ln(1-x)}{x} + 1}{x^{-\alpha}} \right) = \lim_{x \rightarrow 0} \left(\frac{-\frac{x}{2} - \frac{x^2}{3} \dots}{x^{-\alpha}} \right)$$

For limit to exist $\alpha = -1$

$$l = -\frac{1}{2}$$

$$19.(B) \quad \int \frac{a^{\sqrt{x}}}{\sqrt{x}} dx$$

$$\text{Substituting } \sqrt{x} = t$$

$$\frac{1}{2} \cdot \frac{1}{\sqrt{x}} dx = dt$$

$$= \int a^{\sqrt{x}} \left(\frac{1}{2\sqrt{x}} dx \right) = 2 \int a^t dt = \frac{2a^t}{\log_e a} + c = 2a^{\sqrt{x}} \log_a e + c$$

20.(C) Let cylinder has radius r and height h .

$\triangle AO'B$ and $\triangle AOC$ are similar

$$\text{So, } \frac{AO'}{AO} = \frac{O'B}{OC}$$

$$\Rightarrow \frac{H-h}{H} = \frac{r}{R} \Rightarrow 1 - \frac{h}{H} = \frac{r}{R}$$

$$\Rightarrow \frac{h}{H} = 1 - \frac{r}{R} \Rightarrow h = H \left(1 - \frac{r}{R} \right)$$

$$V = \pi r^2 h$$

$$V(r) = \pi r^2 \cdot H \left(1 - \frac{r}{R} \right)$$

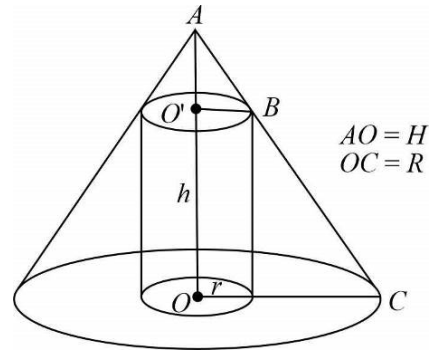
$$V(r) = \pi H \left(r^2 - \frac{r^3}{R} \right)$$

For maximum volume

$$V'(r) = \pi H \left(2r - \frac{3r^2}{R} \right) = 0$$

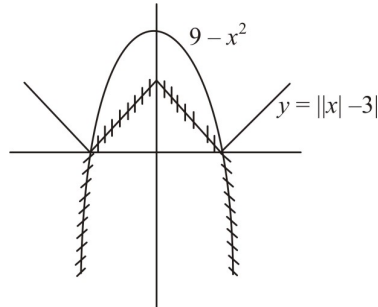
$$\Rightarrow 2r - \frac{3r^2}{R} = 0 \Rightarrow 2 - \frac{3r}{R} = 0 \quad (r \neq 0)$$

$$\Rightarrow r = \frac{2R}{3}$$



SECTION-2

21.(3) It is clear from the graph that there are three points where there is sharp corner. Hence there are three points where function is not differentiable.



22.(0) $f(x) = f(1-x)$

$$\Rightarrow f'(x) = -f'(1-x)$$

$$\text{at } x = \frac{1}{2}$$

$$f'\left(\frac{1}{2}\right) = -f'\left(1 - \frac{1}{2}\right)$$

$$f'\left(\frac{1}{2}\right) = -f'\left(\frac{1}{2}\right)$$

$$\Rightarrow f'\left(\frac{1}{2}\right) = 0 \quad \dots\dots\dots(i)$$

Also $x = \frac{1}{4}$

$$f'\left(\frac{1}{4}\right) = -f'\left(\frac{3}{4}\right)$$

$$\Rightarrow f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right) = 0 \quad \dots\dots\dots(ii)$$

By (i) and (ii)

$$f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right) = 0$$

23.(4.00) The value of n is the least degree of numerator.

$$\begin{aligned} \text{Nr. } (\cos x - 1)(\cos x - e^x) - \frac{x^3}{2} \text{ (expanding this)} \\ = \left(1 - \frac{x^2}{2} + \frac{x^4}{24} \dots - 1\right) \left(\left(1 - \frac{x^2}{2} + \frac{x^4}{24} \dots\right) - \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} \dots\right)\right) - \frac{x^3}{2} \\ = \left(-\frac{x^2}{2} + \frac{x^4}{24} \dots\right) \left(-x - x^2 - \frac{x^3}{6} - \frac{x^5}{120} \dots\right) - \frac{x^3}{2} \\ = \frac{x^3}{2} - \frac{x^5}{24} \dots + \frac{x^4}{2} - \frac{x^6}{24} \dots - \frac{x^3}{2} \end{aligned}$$

So, minimum degree of numerator is 4.

So, the value of n is 4.

24.(2.50) $\frac{f(3) - f(2)}{3 - 2} = f'(\alpha)$

$$f(3) \leq 5 \quad \dots(i)$$

$$\frac{f(5) - f(3)}{5 - 3} = f'(\beta)$$

$$f(3) \geq 5 \quad \dots(ii)$$

Hence $f(3) = 5$

25.(4.00)

$$f(x) = \begin{cases} 2 & -3 < x < -2 \\ 1 & -2 \leq x < -1 \\ 0 & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ 1 & 1 \leq x < 2 \\ 2 & 2 \leq x < 3 \end{cases}$$

Which is discontinuous at $x = -2, -1, 1, 2$